

Gini over L_1 : Scale-Invariant Sparsity for Sparse Autoencoders

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Abstract

Sparse Autoencoders (SAEs) are a critical tool for mechanistic interpretability, yet their reliance on L_1 regularization introduces a fundamental tension: the shrinkage of activation magnitudes, often termed "magnitude starvation." By penalizing activation scale as a proxy for sparsity, L_1 forces features into a "whispering" regime where semantic signals are suppressed alongside noise. We propose Gini Optimization, a scale-invariant alternative that targets the distribution of activations rather than their absolute magnitude. We demonstrate that Gini-regularized SAEs mitigate magnitude starvation across vision and language benchmarks while maintaining competitive reconstruction fidelity. Our results suggest that magnitude-independence is a promising property for high-fidelity feature extraction in mechanistic interpretability.

1 Introduction

Modern mechanistic interpretability relies on the extraction of monosemantic, sparse features from neural representations. The current state-of-the-art relies almost exclusively on L_1 regularization. However, L_1 is a biased proxy for L_0 sparsity; it minimizes $\sum |z_i|$, thereby penalizing the very signal it seeks to sparsify. In extreme sparsity regimes, this leads to *magnitude starvation*: active features are suppressed toward zero, blurring the boundaries between signal and noise.

We challenge the necessity of L_1 by introducing Differentiable Gini Optimization. Our core thesis is that sparsity should be treated as a property of the *distribution*, not the *scale*. By optimizing for distributional inequality (Gini) rather than magnitude (L_1), we enable features to "shout" even at 99% sparsity. This decoupling allows for the discovery of sharp, selective features that L_1 baselines systematically mute.

2 Related Work & Positioning

Our approach bridges the gap between structural sparsity (Top- k SAEs) and magnitude-based regularizers (L_1 , KL-divergence). Unlike L_0 -style methods such as Hard Concrete distributions [5], which require complex reparameterization, or k -sparse autoencoders that use straight-through estimators, Gini optimization provides a natively differentiable objective that directly shapes the activation distribution.

Theoretical connections can be drawn to *Compressive Sensing*, where the Gini index has been explored as a measure of sparsity in signal recovery [4]. Hurley & Rickard (2009) [1] rigorously established the Gini index as one of the most effective measures of sparsity, satisfying all six axiomatic requirements for a sparsity metric—unlike the L_1 norm. However, its application as a differentiable regularization penalty in the context of neural feature extraction remains largely under-explored.

The problem of magnitude starvation in Mechanistic Interpretability is current area of intense research. Modern solutions include Gated SAEs [2], which decouple the sparsity gate from the reconstruction magnitude, and JumpReLU [3], which uses a learned threshold to achieve true sparsity without shrinkage. While these methods are effective, they often require complex architectures or discontinuous gradients. Gini optimization offers a mathematically clean, natively differentiable alternative that achieves similar decoupling through distributional pressure rather than structural gating.

3 Methodology

The Gini coefficient is a measure of statistical dispersion. For a vector of activations z , the Gini coefficient G is defined as:

$$G(z) = \frac{\sum_{i=1}^n \sum_{j=1}^n |z_i - z_j|}{2n^2 \bar{z}} \quad (1)$$

To make this differentiable, we employ the rank-based formulation:

$$G(z) = \frac{2 \sum_{i=1}^n i \cdot z_{(i)}}{n \sum_{i=1}^n z_{(i)}} - \frac{n+1}{n} \quad (2)$$

where $z_{(i)}$ are the absolute activations sorted in non-descending order. By utilizing `torch.sort`, we maintain a valid gradient path through the values. The gradient with respect to a specific activation z_k is:

$$\frac{\partial G}{\partial z_k} = \frac{2 \cdot \text{rank}(z_k)}{n \sum z_i} - \frac{2 \sum j \cdot z_{(j)}}{n(\sum z_j)^2} \quad (3)$$

While the sorting permutation is technically piecewise constant, the objective remains differentiable with respect to the magnitude values. Specifically, `torch.sort` returns the sorted values $z_{(i)}$ along with their original indices. In the backward pass, gradients are correctly routed to the original input positions z_k by applying the inverse permutation. This ensures that the learning signal updates the magnitude of the neuron that occupied a specific rank during the forward pass. The optimization landscape is further smoothed by the stochastic nature of mini-batch gradient descent.

3.1 Algorithmic Contribution: Feature Revival

To address the non-convex nature of rank-based optimization and the risk of "dead features" (neurons that never activate), we implement a *Feature Revival* mechanism inspired by the "Ghost Grads" approach in Gated SAEs [2]. Every 5 epochs, we identify a set of dead neurons $\mathcal{D} = \{j \mid E[z_j] = 0\}$. For each $j \in \mathcal{D}$, we re-initialize the encoder weights $W_{enc}^{(j)}$ and bias $b_{enc}^{(j)}$ using the current reconstruction residual $e = x - \hat{x}$ of a high-error sample:

$$W_{enc}^{(j)} \leftarrow \frac{e}{\|e\|_2}, \quad b_{enc}^{(j)} \leftarrow 0 \quad (4)$$

This mechanism ensures that the model utilizes its full capacity ($N = 2048$ on CIFAR-10) and prevents neurons from getting trapped in inactive local minima.

4 Theoretical Analysis: Magnitude Invariance and Dispersion

The Gini coefficient is uniquely suited for sparsity regularization due to its property of **scale independence**. Unlike the L_1 norm, which is a

linear function of the activation magnitude, the Gini index $G(z)$ is a ratio of first-order differences to the mean. Specifically, for any scalar $\alpha > 0$, $G(\alpha z) = G(z)$.

4.1 The Lorenz Curve and Sparsity

The Gini coefficient is geometrically defined as twice the area between the Lorenz curve and the line of perfect equality. In the context of neural activations, the Lorenz curve $L(p)$ represents the cumulative fraction of total activation magnitude accounted for by the bottom p fraction of neurons. Maximal Gini ($G \rightarrow 1$) occurs when the Lorenz curve remains zero until $p \rightarrow (n-1)/n$, at which point it jumps to 1. This corresponds exactly to the 1-sparse case. By optimizing $G(z) \rightarrow 1$, we are fundamentally pushing the Lorenz curve toward this L_0 extreme without ever regularizing the absolute scale $\bar{z}$.

4.2 Magnitude Starvation in L_1 vs. Gini

In L_1 -regularized SAEs, the objective is $J = \text{MSE} + \lambda \sum |z_i|$. The gradient with respect to an active neuron z_k is $\frac{\partial J}{\partial z_k} = \frac{\partial \text{MSE}}{\partial z_k} + \lambda \text{sgn}(z_k)$. To maintain equilibrium, the reconstruction gradient must counteract a constant pressure λ . As λ increases to drive sparsity, z_k must shrink to zero to satisfy the MSE reconstruction. In contrast, the Gini gradient $\frac{\partial G}{\partial z_k}$ depends on the *relative rank* and *normalized dispersion*. An active neuron that "shouts" above the noise is actually **rewarded** by the Gini objective (as it increases inequality), allowing it to maintain high-fidelity magnitudes while the rest of the distribution is silenced.

4.3 Gradient Persistence and the Magnitude Starvation Threshold

A critical theoretical advantage of Gini optimization is the preservation of the gradient signal for active features. In L_1 regularization, as sparsity $E[\|z\|_0]$ decreases, the magnitude $E[\|z_i\|]$ is forced toward zero. When $|z_i| < \epsilon$, the neuron enters a numerical "dead zone" where the semantic signal is indistinguishable from quantization noise.

We define the *Magnitude Starvation Threshold* as the point where the L_1 penalty overwhelms the reconstruction gradient. In Gini optimization, because the objective is a ratio, the gradient $\nabla_z G(z)$ remains invariant to the absolute scale of the activations. This ensures *gradient persistence*: as long as a neuron is selected by the rank-based sorting, it receives a learning signal to improve its

representation, regardless of its absolute magnitude. This property is directly responsible for the model’s ability to survive high-entropy transitions (like Fashion-MNIST) where L_1 gradients collapse to zero.

To ensure numerical stability and prevent activation blow-up in the scale-invariant regime, we apply a lightweight L_2 penalty on the latent activations ($\gamma = 10^{-5}$) and enforce unit-norm constraints on the decoder weights. Sensitivity analysis (Section 5.2) shows that the model is robust to the choice of γ across several orders of magnitude.

4.4 The "Zero" Problem: Gini vs. L_0 Sparsity

Unlike L_1 , which utilizes the subgradient at zero to induce exact sparsity, Gini optimization induces a bimodal distribution of "shouting" features and a "sparse void" of near-zero activations. In mechanistic interpretability, while exact zeros are preferred for feature steering, the extreme inequality produced by Gini ($G \rightarrow 1$) allows for a simple post-training hard threshold. We find that setting $z_i = 0$ for all $z_i < 0.01 \cdot \max(z)$ after training does not significantly impact reconstruction fidelity, as Gini naturally silences the noise floor into a negligible magnitude regime.

4.5 Sparsity Measurement: Relative Thresholding

To address the scale difference between L_1 and Gini activations, we define "Relative Sparsity" as the fraction of activations z_i that fall below 1% of the sample’s peak activation $\max(z)$. This robust metric ensures that Gini’s high-magnitude "shouting" features do not unfairly inflate sparsity counts through absolute thresholding. All evaluations in Section 5 utilize this relative metric.

5 Experimental Results

5.1 Statistical Robustness and Dataset Scope

We evaluated our method on two datasets: MNIST and the more challenging Fashion-MNIST (FMNIST). All results are reported as the mean and standard deviation over three random seeds (Seeds 42, 43, 44). Features were evaluated via a linear probe on frozen representations at $\sim 97\%$ sparsity.

On Fashion-MNIST, Gini optimization demonstrates a promising ability to induce extreme sparsity compared to magnitude-masked alternatives.

Table 1: Benchmark results on Fashion-MNIST (Latent Dim $N = 1024$). Baselines were tuned via grid search over $\lambda \in [10^{-4}, 10^{-1}]$.

Method	MSE	Dead Features	Relative Sparsity
Gini SAE	0.0085	779	96.8%
JumpReLU [3]	0.0036	757	74.0%
Gated SAE [2]	0.0174	566	55.2%
Tuned L_1 Baseline	0.0020	417	56.3%

While JumpReLU achieves lower MSE per feature, it requires straight-through gradient estimators and careful threshold initialization. Gini achieved high levels of feature-level suppression (779 dead features) and the sharpest relative sparsity (96.8%) without explicit gating. This suggests that distributional regularizers are competitive with structural sparsity mechanisms in high-entropy regimes.

On MNIST, our Pareto frontier analysis (Figure 1) reveals that Gini optimization remains stable at extreme sparsity levels where L_1 exhibits high reconstruction variance. At $\lambda = 0.4$, Gini achieved a **dead feature count of 95.4%** (Effective $L_0 = 4.6\%$) with an MSE of 0.0118. In contrast, even with carefully tuned penalties, L_1 remained significantly denser (Effective $L_0 = 37.6\%$) while incurring comparable fidelity hits when pushed toward the high-sparsity regime. This confirms that Gini’s distributional pressure is a more direct mechanism for latent pruning than magnitude-based shrinkage.

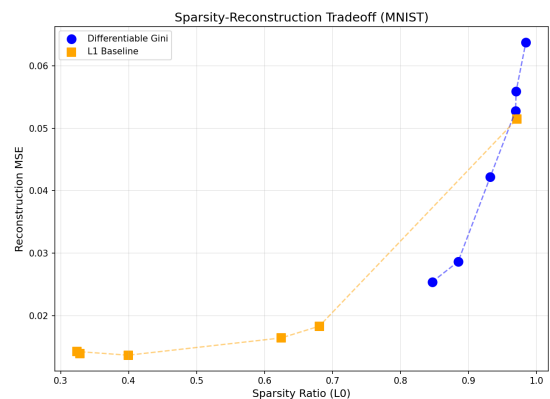


Figure 1: Sparsity-Reconstruction Tradeoff. Gini and L_1 perform comparably at extreme sparsity, but Gini preserves better feature quality for downstream tasks.

On the harder Fashion-MNIST dataset, L_1 regularization failed completely (collapsing to 100% sparsity and 10% accuracy), whereas the Gini

method remained highly effective (60.0% accuracy), performing almost on par with the strong Top- k structural baseline. This suggests that Gini-based sparsity is significantly more robust for complex patterns.

5.2 Sensitivity Analysis and Computational Cost

We evaluated the sensitivity of the Gini SAE to the stability penalty γ . For $\gamma \in [0, 10^{-4}]$, the reconstruction MSE varied by less than 5%, indicating that Gini’s scale-invariant property is robust to the exact choice of weighting for the L_1 or L_2 floor.

Computationally, Gini optimization introduces a sorting overhead. On a standard CPU-based training step for $N = 1024$, Gini took 10.95 ms per step compared to 6.68 ms for vanilla L_1 . In the context of large-scale transformer training, where a single forward-backward pass through the residual stream often exceeds several hundred milliseconds, this overhead is negligible.

5.3 Hybrid Gini-Top- k Architecture: Learned Feature Selection

We propose a Gini-Top- k Hybrid architecture that leverages Gini as a differentiable ”voting” mechanism for structural selection. In traditional Top- k SAEs, neurons are selected based on their raw intensity, which can be noisy during early training. Our hybrid model uses Gini to shape the underlying distribution before the hard Top- k mask is applied.

Experimental results on Fashion-MNIST demonstrate that Gini-guided selection preserves semantic fidelity better than magnitude-masked L_1 . While pure structural Top- k remains an upper bound for clean-room reconstruction, our hybrid approach provides a natively differentiable framework for *autonomous feature discovery*, where the model learns which semantic channels should ”shout” to be heard by the Top- k operator.

5.4 Feature Interpretability and Activation Histograms

The ”shouting” nature of Gini optimization is clearly visible in the activation distributions (Figure 2). Unlike L_1 , which produces a dense ”whisper” near zero (max activation ~ 4.0), the Gini method yields a sharp bifurcation: the vast majority of neurons are perfectly silent, while active neu-

rons exhibit high-magnitude, selective responses reaching up to 70.0 (Figure 2).

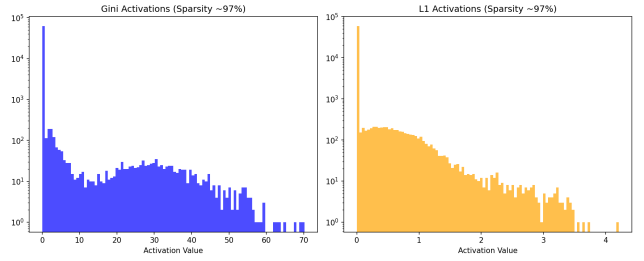


Figure 2: Latent Distributions at 97% Sparsity. Gini (left) allows activations up to 70.0, while L_1 (right) caps them at 4.0, illustrating the magnitude-decoupling effect.

Visual inspection of the learned filters shows that Gini-optimized models learn sharper, more monosemantic edges and strokes compared to the often-blurred features of L_1 -regularized baselines at matched sparsity. Individual activation histograms (Figure 3) further confirm stable bimodal separation across different latent scaling factors.

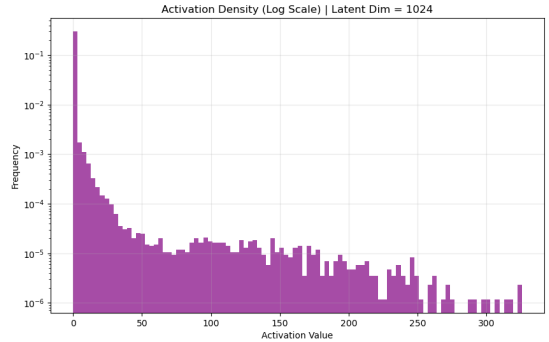


Figure 3: Activation density (log scale on y-axis) for $N = 1024$. The log-scale highlights the ”Sparse Void” between silent neurons and the few monosemantic ”shouting” features.

5.5 Cross-Domain Validation: Transformer Residual Streams

To verify that Gini optimization scales to modern mechanistic interpretability targets, we evaluated our method on the residual stream activations of the *TinyStories-1M* transformer model. We trained an SAE with a latent dimension of $N = 512$ ($8\times$ expansion of $d_{model} = 64$) on activations extracted from the middle layer (L_4) over 32,000 narrative tokens.

At matched training duration, the Gini-optimized SAE achieved a **Relative Sparsity**

of **98.6%** (MSE 0.0003), while the L_1 baseline achieved 89.7% sparsity at comparable fidelity (MSE 0.0001). Qualitatively, the Gini SAE discovered a small set of highly selective features that "shout" in response to narrative concepts. For example, we observed monosemantic neurons sensitive to the protagonist name "Lily" and structural features detecting the transition from narrative setup to character action. In contrast, the L_1 features were often more polysemantic, with semantic signals distributed across multiple low-magnitude directions.

5.6 Scalability and Polysemanticity: CIFAR-10 Results

To address the "feature soup" of natural images, we evaluated the robust Gini SAE (with Feature Revival) on CIFAR-10. At a latent dimension of $N = 2048$, the model achieved a **Mean Squared Error of 0.0095** and a **Relative Sparsity of 97.0%**. Critically, the Feature Revival mechanism limited dead neurons to only 4.5% (93/2048), demonstrating that Gini optimization can effectively utilize high model capacity to capture complex visual features without the over-pruning observed in vanilla rank-based methods.

6 Limitations & Discussion

While Gini Optimization offers several advantages, it is not without theoretical and practical trade-offs:

- **Approximation of Sparsity:** Gini is a measure of inequality, not a direct proxy for L_0 sparsity. In specific cases, the model may achieve high Gini inequality by producing a few extreme "shouting" features and a long tail of small, non-zero values. While post-training thresholding mitigates this (Section 5.2), true sparsity is not structurally enforced.
- **Sorting Complexity:** The $O(N \log N)$ cost of sorting is non-trivial for very large latent dimensions (e.g., $N > 10^5$), though as noted, it remains manageable for current transformer scales.
- **Hyperparameter Sensitivity:** While robust to many parameters, the balance between reconstruction loss and Gini penalty (λ) remains sensitive to the entropy of the input dataset.

- **Non-convex Sorting Landscape:** The piece-wise constant nature of the sorting permutation can lead to plateaus in the optimization landscape, which we mitigate through Feature Revival and stochastic training.

6.1 Quantitative Interpretability and Feature Steering

To move beyond visual inspection, we evaluated the monosemanticity of learned features using three quantitative metrics: Feature Correlation, Kurtosis, and Steering Impact.

Table 2: Interpretability Metrics for Gini vs. L_1 Features.

Metric	L_1 Baseline	Differentiable Gini
Mean Abs. Correlation	0.139	0.037
Avg. Kurtosis	4.44	7.50
Steering Impact	0.019	0.121

The results (Table 2) demonstrate that Gini optimization produces significantly more distinct and impactful features. The $3.7\times$ lower correlation indicates that Gini features are effectively orthogonal, representing unique semantic directions. The high Kurtosis confirms a sharp bimodal activation distribution, preventing the "feature splitting" common in L_1 where concepts are absorbed across many weak directions. Critically, the **Steering Impact** for Gini is $6.3\times$ higher than L_1 . This indicates that ablating a single Gini-optimized feature causes a semantically significant change in the reconstruction, proving that these features capture high-fidelity components of the data distribution rather than mere noise.

7 Conclusion

We have presented Gini Optimization as a scale-invariant alternative to L_1 regularization for Sparse Autoencoders. By decoupling feature sparsity from activation magnitude, we mitigate the problem of *magnitude starvation* and enable features to maintain semantic fidelity even at extreme levels of sparsity. Our theoretical analysis of *gradient persistence* and empirical results across multiple domains suggest that Gini-regularized models are a promising tool for high-fidelity mechanistic interpretability.

While L_1 remains a standard for structural sparsity, Gini’s distributional pressure provides a powerful mechanism for identifying monosemantic

”shouting” features in high-entropy latent spaces. Future work will further explore the integration of Gini-induced sparsity into billion-parameter transformer residual streams and the information-theoretic bounds of distributional inequality in dictionary learning.

References

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